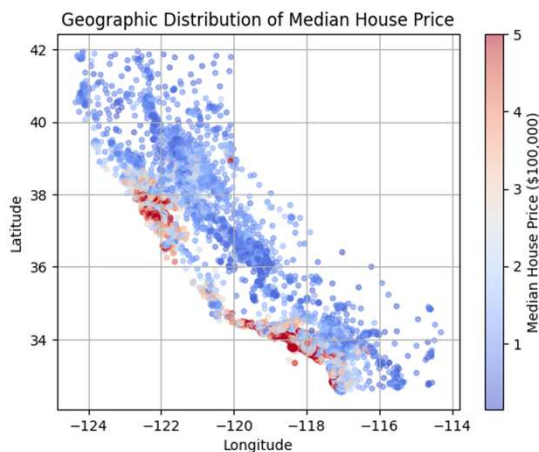


Predicting California Housing Values with Ridge Regression and PyTorch Neural Networks

Project Question: Can a PyTorch neural network improve California median house value prediction compared with Ridge Regression, and how important are geographic features such as latitude and longitude for predictive performance?

Dataset

- California Housing Dataset from scikit-learn
- 20,640 census block groups
- 8 features: income, house age, rooms, bedrooms, population, occupancy, latitude, longitude
- Target: median house value in \$100,000 units

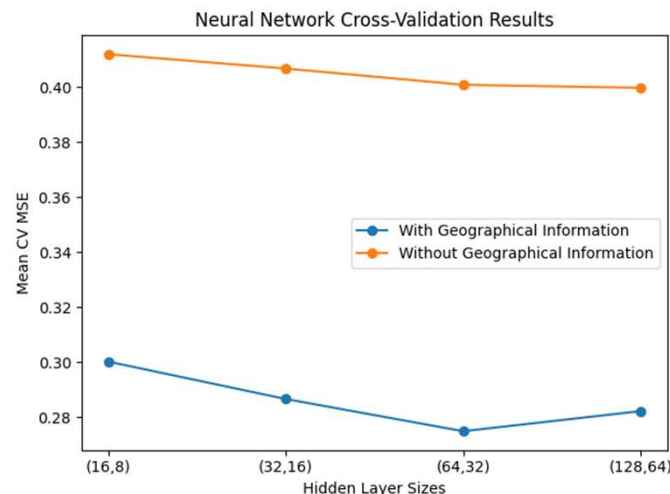


Machine Learning Methods

- Ridge Regression baseline
- Two-layer PyTorch neural network
- Both models trained with and without latitude/longitude

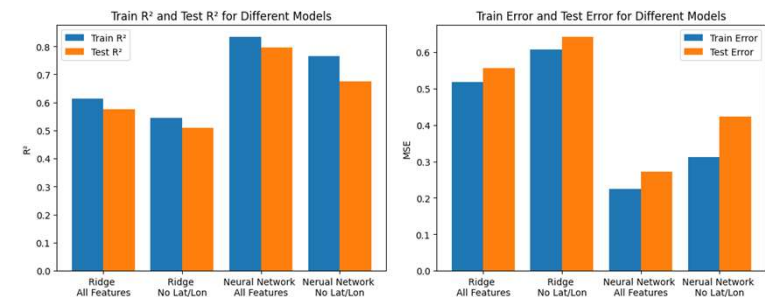
Cross-validation and Model Selection

- Ridge: 5-fold CV over $\alpha = [0.01, 0.1, 1, 10, 100]$
- Best α : 0.01 with geo, 10 without geo
- Ridge CV MSE was nearly flat across α values
- NN: 5-fold CV over hidden sizes: (16,8), (32,16), (64,32), (128,64)
- Best NN: (64,32) with geo, (128,64) without geo



Model Comparison Results

- Best Test Results:
 - Best model: NN with all features
 - Test MSE = 0.2686
 - Test $R^2 = 0.7950$
 - Ridge baseline: MSE = 0.5559, $R^2 = 0.6126$



Conclusions

- Best model: NN with all features
- Removing latitude/longitude reduced performance
- Geographic features are important for housing prediction

Future Work

- Compare with tree-based models
- Tune more NN hyperparameters

